Beam-Path Heterogeneity Metrics

Correlation with GPR & Energy-Stratified Analysis ($n = 3\,950$)

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Recap & New Approach

Last week: spherical neighbourhood around BP $\rightarrow \sigma_{\mathrm{HU}}$, TV, CV as heterogeneity metrics.

Limitation: a single sphere ignores *where* along the beam path the density transitions occur.

This week: extended pipeline

1. **Density region analysis**: segment beam path into contiguous tissue regions inside the BP range.
2. **Six advanced heterogeneity metrics**: max HU jump, σ_{HU} at BP, max HU gradient, lateral HU variance, heterogeneity fraction, interface–BP distance.
3. **Two 3-D Sobel edge metrics**: mean axial Sobel magnitude and 95th-percentile Sobel in the BP region.
4. **Energy-stratified correlation analysis**: binned correlations, energy-coloured scatter plots, and partial correlations controlling for beam energy.

Dataset: $n = 3\,950$ beamlets from the ADoTA training set, beam energies 70 MeV to 250 MeV.

Tissue Segmentation: HU Look-Up Table

Each CT voxel is assigned to one of **7 tissue classes** based on its Hounsfield Unit value.

Class	Label	HU range	Colour
0	Air	$[-1024, -950)$	black
1	Lung	$[-950, -500)$	dark blue
2	Fat	$[-500, -100)$	cyan
3	Soft tissue	$[-100, 100)$	green
4	Contrast/blood	$[100, 300)$	yellow
5	Cancellous bone	$[300, 500)$	orange
6	Cortical bone	$[500, 3072)$	red

Implementation: `segment_hu()` iterates over the LUT and assigns the class index whose HU interval contains the voxel value.

Boundaries derived from the Hounsfield scale reference values (`https://en.wikipedia.org/wiki/Hounsfield_scale`) and verified against clinical tissue classification ranges.

Gaussian Smoothing of CT Before Segmentation

Problem: raw CT volumes contain acquisition noise. Direct segmentation on raw data can produce noisy class maps with many spurious small connected regions, inflating the density region count.

Solution: apply a 3-D Gaussian filter before segmentation. Conceptually,

$$\tilde{I}(\mathbf{r}) = \big(G_\sigma * I\big)(\mathbf{r}), \qquad G_\sigma(\mathbf{r}) = \frac{1}{(2\pi\sigma^2)^{3/2}} \exp\!\Big(-\frac{|\mathbf{r}|^2}{2\sigma^2}\Big)$$

On the voxel grid, this is implemented numerically as a discrete separable convolution along each axis.

- Implementation: `from scipy.ndimage import gaussian_filter`
  `ct_hu_smooth = gaussian_filter(ct_hu, sigma=1.0, mode='reflect')`
- $\sigma = 1.0$ voxel $= 2\,\text{mm}$ for a $2 \times 2 \times 2\,\text{mm}$ grid.
- Effect: suppresses high-frequency noise and reduces spurious small regions before segmentation.
- Result: density region count better reflects anatomical structure rather than voxel-scale noise.

Bragg Peak Range from Ground-Truth IDD

Integrated Depth Dose (IDD): collapse the 3-D dose to 1-D:

$$\mathrm{IDD}[k] \;=\; \sum_{y}\sum_{x} D[k,y,x], \qquad k = 0,\ldots,N_z - 1$$

Bragg peak index: $k^* = \arg\max_k \; \mathrm{IDD}[k]$

Proximal boundary z_{min}: walk *backward* from k^* towards the entrance:

$$z_{\mathrm{min}} = \max\left\{\, k \le k^* \;\middle|\; \mathrm{IDD}[k] < 0.50 \cdot \mathrm{IDD}_{\mathrm{max}} \,\right\}$$

Anchored to the peak, robust against entrance plateau noise. Analogous to clinical proximal R50.

Distal boundary z_{max}: walk *forward* from k^*:

$$z_{\mathrm{max}} = \min\left\{\, k \ge k^* \;\middle|\; \mathrm{IDD}[k] < 0.10 \cdot \mathrm{IDD}_{\mathrm{max}} \,\right\}$$

Marks the distal fall-off at 10% of peak dose.

Physical units: $z_{\mathrm{min}\,/\,\mathrm{max}}[\mathrm{mm}] = z_{\mathrm{min}\,/\,\mathrm{max}}[\mathrm{slices}] \times 2\,\mathrm{mm/slice}$.

Density Region Analysis: Overview

Goal: characterise *how many tissue transitions* the beam traverses inside the Bragg peak zone.

Steps:

1. For each depth slice $k \in [z_{\min}, z_{\max}]$, compute the **flux-weighted mean HU**:

$$\bar{H}[k] = \frac{\sum_{y,x} w_{k,y,x} \, I_{k,y,x}}{\sum_{y,x} w_{k,y,x}}$$

   where $w_{k,y,x} = |\mathrm{flux}[k,y,x]|$ if $\geq 0.10 \cdot \max_{y,x} |\mathrm{flux}[k]|$, else 0.

2. Segment $\bar{H}[k]$ into tissue class $c[k]$ using the HU LUT.

3. Group consecutive slices with the **same class** into contiguous **density regions**.

Flux-weighted $\bar{H}[k]$ $\downarrow$ Segment $\rightarrow c[k]$ $\downarrow$ Group regions $\downarrow$ n_{regions}, ΔHU

Density Region Analysis: Metrics

Two output metrics per beamlet:

1. Number of density regions (n_{regions}):
Count of maximal contiguous groups of slices sharing the same tissue class along the beam path inside $[z_{\min}, z_{\max}]$.

Example: Soft tissue $\rightarrow$ Bone $\rightarrow$ Soft tissue $=$ 3 regions.

2. Total HU change (Δ_{total}):
Sum of absolute mean-HU differences between consecutive regions:

$$\Delta_{\text{total}} = \sum_{j=1}^{n_{\text{regions}}-1} \left| \bar{H}_j - \bar{H}_{j-1} \right|$$

where $\bar{H}_j$ is the mean HU of region j.

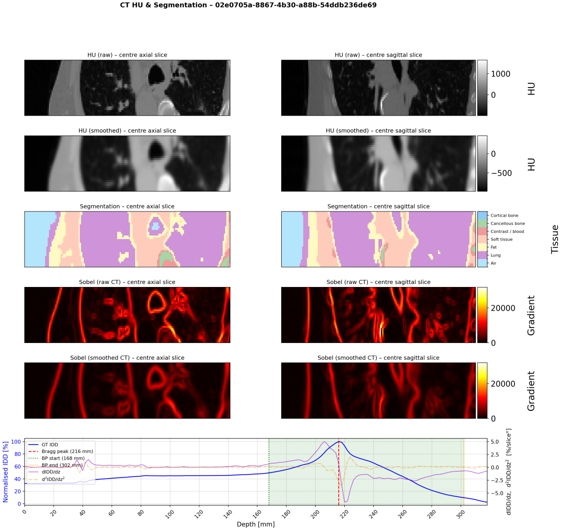
Hypothesis: higher n_{regions} and/or Δ_{total} $\longrightarrow$ harder sample $\longrightarrow$ lower GPR.

Density Region Analysis: Pseudocode

Input: CT volume $I \in \mathbb{R}^{D\times H\times W}$, flux $F \in \mathbb{R}^{D\times H\times W}$, BP range $[z_{\min}, z_{\max}]$
Output: n_{regions}, Δ_{total}

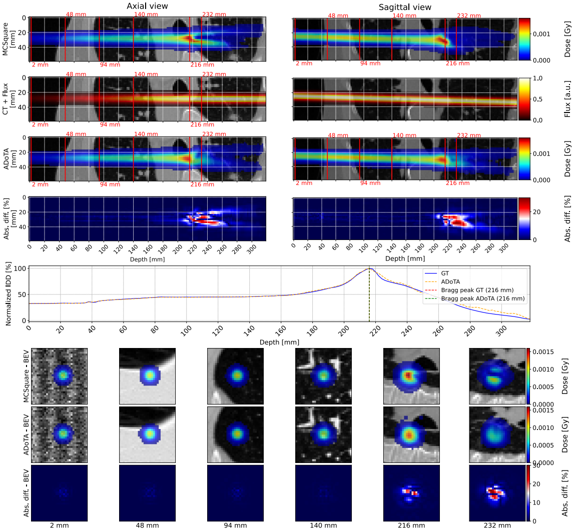
for $k \leftarrow \lceil z_{\min} \rceil$ **to** $\lfloor z_{\max} \rfloor$ **do**
  $f_{\max} \leftarrow \max_{y,x} |F[k,y,x]|$
  $\text{mask} \leftarrow |F[k]| \geq 0.10 \cdot f_{\max}$
  $\bar{H}[k] \leftarrow \text{weighted\_avg}\big(I[k,\text{mask}],\ |F[k,\text{mask}]|\big)$
end
$c[k] \leftarrow \textsc{SegmentHU}\big(\bar{H}[k]\big)$ `// tissue class per slice`
Group consecutive slices with same $c[k]$ into regions $R_0, R_1, \ldots$
$n_{\text{regions}} \leftarrow$ number of regions
$\Delta_{\text{total}} \leftarrow \sum_{j=1}^{n_{\text{regions}}-1} |\bar{H}_j - \bar{H}_{j-1}|$
return $n_{regions}$, Δ_{total}

Example Beamlet: CT Segmentation, Sobel Edges & IDD



Beamlet 02e0705a, 127.55 MeV. Row 1: raw CT. Row 2: Gaussian-smoothed CT (σ=1 vox). Row 3: 7-class segmentation. Row 4: raw Sobel magnitude. Row 5: smoothed Sobel magnitude. Bottom: IDD with BP range (green:

Example Beamlet: ADoTA Dose Prediction



Same beamlet. Top: CT input & beamlet shape (flux). Bottom: ground-truth dose (MC), ADoTA prediction, and gamma index map (2%/2 mm). Depth slices sampled uniformly from entrance to distal fall-off.

Advanced Heterogeneity Metrics

Six additional metrics computed along the beam path in the BP range:

Metric	Symbol	Definition
Max HU jump	$\Delta H_{\max}$	Largest $\lvert \bar{H}_j - \bar{H}_{j-1} \rvert$ between consecutive regions
σ_{HU} at BP	σ_{BP}	Std. dev. of flux-weighted mean HU along BP slices
Max HU gradient	$g_{\max}$	$\max_k \lvert \bar{H}[k+1] - \bar{H}[k] \rvert$ (slice-to-slice)
Lateral HU variance	$\mathrm{Var}_{\perp}$	Mean intra-slice HU variance within flux mask
Heterogeneity fraction	f_{het}	Fraction of BP slices with tissue class $\neq$ dominant class
Interface–BP distance	d_{int}	Distance (mm) from nearest tissue interface to k^*

Rationale: these metrics capture complementary aspects of tissue complexity: magnitude of transitions ($\Delta H_{\max}$, $g_{\max}$), overall variability (σ_{BP}, $\mathrm{Var}_{\perp}$), spatial extent of heterogeneity (f_{het}), and proximity of interfaces to the dose-sensitive BP region (d_{int}).

3-D Sobel Edge Detection Metrics

Motivation: Sobel operators detect tissue boundaries directly from the CT gradient, without requiring segmentation.

Method:

1. Smooth CT with Gaussian ($\sigma = 1.0$ voxel) to suppress noise.
2. Apply 3-D Sobel filters along depth (z), height (y), and width (x) axes:

$$S(\mathbf{r}) = \sqrt{S_z^2 + S_y^2 + S_x^2}$$

3. Aggregate within the flux-weighted beam path.

Two output metrics:

- **Mean axial Sobel** ($\bar{S}_{\mathrm{axial}}$): flux-weighted mean of $|S_z|$ across the full beam path, capturing axial density transitions along the beam direction.
- **P95 Sobel at BP** ($S_{95,\mathrm{BP}}$): 95th percentile of $S(\mathbf{r})$ within the BP range, weighted by flux, capturing the most extreme edge responses near the Bragg peak.

Dataset & Model Inference

Dataset:

- $n = 3\,950$ beamlets from the ADoTA training set
- Energy range: 70 MeV to 250 MeV
- Grid: $160 \times 30 \times 30$ voxels, $2 \times 2 \times 2$ mm

Model: ADoTA v3 (DoTA_v3_grid_search_v11)
Gamma pass rate: 2%/2 mm, 10% lower dose cutoff

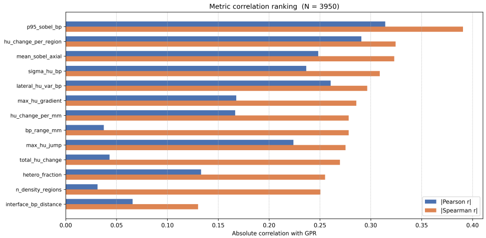
GPR distribution:
Mean $= 98.81\%$, $\sigma = 1.49\%$
Range: [86.97%, 100.00%]

Energy bin distribution:

Bin (MeV)	n	GPR (%)
70–100	892	98.33 ± 2.02
100–130	888	98.77 ± 1.50
130–160	1018	98.90 ± 1.45
160–190	781	99.27 ± 0.77
190–220	195	99.14 ± 0.58
220–250	176	98.59 ± 0.80

Lower energies show larger GPR variance.

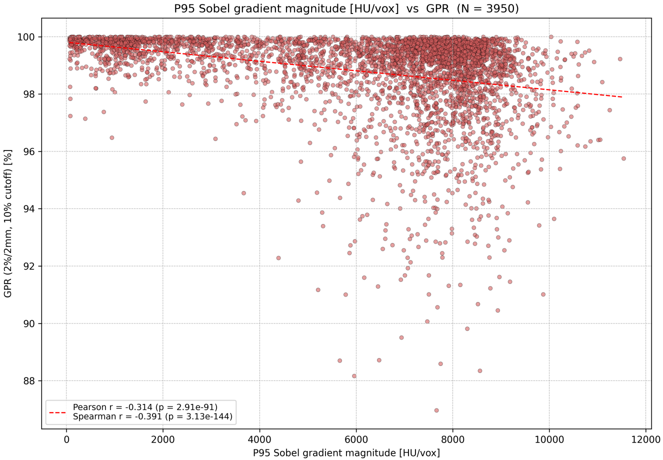
Overall Correlation Ranking ($n = 3\,950$)



Metric	Spearman	Pearson
p95_sobel_bp	-0.391	-0.314
hu_change_per_region	-0.324	-0.291
mean_sobel_axial	-0.323	-0.248
σ_{HU} at BP	-0.309	-0.237
lateral_hu_var_bp	-0.297	-0.261
max_hu_gradient	-0.286	-0.168
hu_change_per_mm	-0.278	-0.167
bp_range_mm	-0.278	-0.037
max_hu_jump	-0.275	-0.224
total_hu_change	-0.270	-0.043
hetero_fraction	-0.255	-0.133
n_{regions}	-0.251	-0.031
interface_bp_dist.	$+0.130$	$+0.066$

All $p < 10^{-5}$. Spearman > Pearson suggests **monotonic but non-linear** relationships.

Best Predictor: P95 Sobel at BP



$S_{95,\mathrm{BP}}$ achieves the strongest correlation with GPR (Spearman $\rho = -0.391$). Higher edge responses in the BP region are associated with lower gamma pass rates.

Motivation for Energy Stratification

Observation: beam energy influences both the anatomy traversed and the model's prediction difficulty.

Question: are the observed correlations genuine reflections of tissue heterogeneity, or are they confounded by energy?

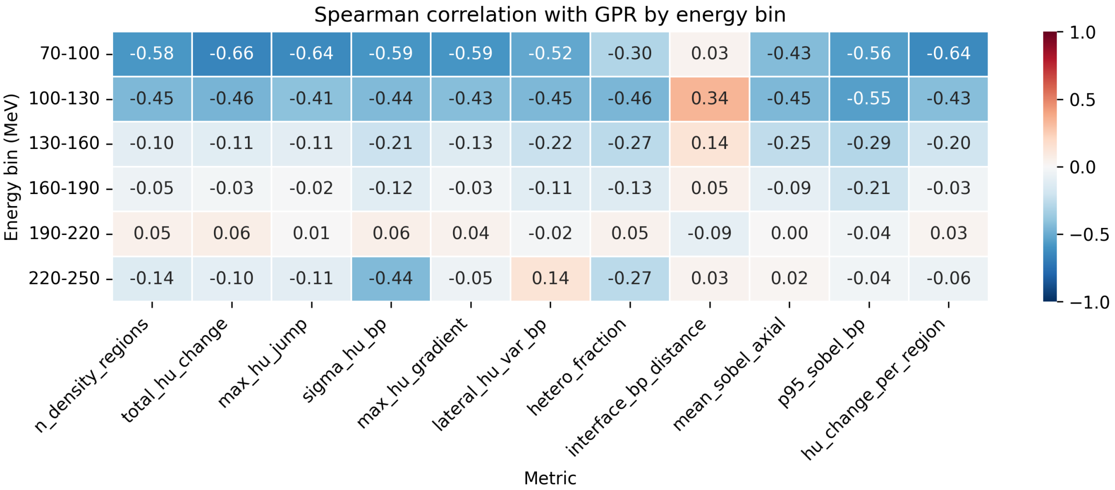
Three complementary analyses:

1. **Binned correlations**: compute Spearman ρ within each 30 MeV energy bin independently.
2. **Energy-coloured scatter plots**: visualise metric vs GPR with colour encoding beam energy.
3. **Partial correlations**: compute partial Spearman correlation controlling for energy:

$$\rho_{XY\cdot Z} = \frac{\rho_{XY} - \rho_{XZ}\,\rho_{YZ}}{\sqrt{(1-\rho_{XZ}^2)(1-\rho_{YZ}^2)}}$$

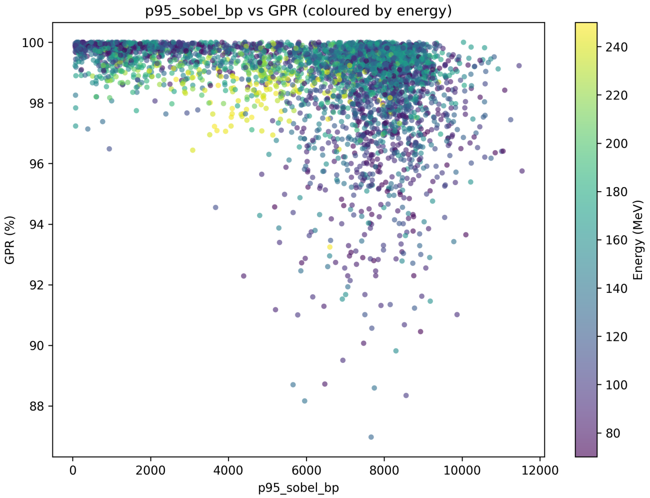
where $X =$ metric, $Y =$ GPR, $Z =$ energy.

Spearman Correlation by Energy Bin



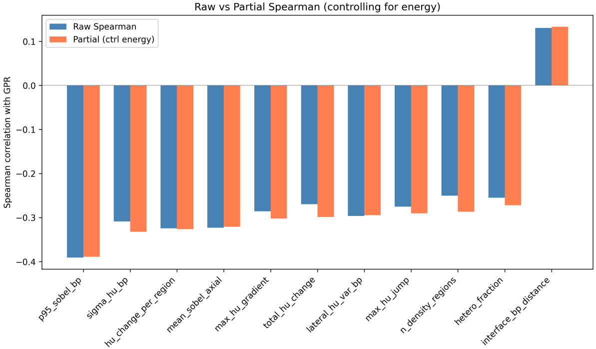
Correlations are **strongest at low energies** (70–130 MeV: $|\rho| > 0.4$ for top metrics), weaken above 160 MeV, and largely vanish in the 190–220 MeV bin.

Energy-Coloured Scatter: P95 Sobel at BP



Low-energy beamlets (dark) populate the high-S_{95} / low-GPR region. At higher energies the metric range compresses and GPR clusters near 99–100%.

Partial Correlations Controlling for Energy



Metric	Raw	Partial
p95_sobel_bp	−0.391	−0.389
$\sigma_{\mathrm{HU,BP}}$	−0.309	−0.332
$\Delta H / n_{\mathrm{reg}}$	−0.324	−0.326
$\bar{S}_{\mathrm{axial}}$	−0.323	−0.321
g_{max}	−0.286	−0.302
n_{regions}	−0.251	−0.287

Raw and partial correlations differ by $|\Delta| < 0.04$: heterogeneity–GPR associations are **not confounded by energy**. For n_{regions} ($\Delta = -0.036$), controlling for energy *strengthens* the correlation.

Key Findings

1. **P95 Sobel at BP is the strongest single predictor** of GPR degradation ($\rho_s = -0.391$), capturing sharp density transitions near the Bragg peak via gradient-based edge detection.
2. **Spearman $\gg$ Pearson** for most metrics, indicating **monotonic but non-linear** relationships between tissue heterogeneity and prediction difficulty.
3. **Correlations are energy-dependent**: strongest at low energies (70–130 MeV, $|\rho_s|$ up to 0.66), weak to absent above 160 MeV.
   - Low-energy beams have shorter ranges $\rightarrow$ tissue heterogeneity has proportionally larger impact.
   - High-energy beams traverse more uniform deep tissue.
4. **Partial correlations confirm**: energy is *not* a confounder: the heterogeneity–GPR association persists after controlling for beam energy ($|\Delta\rho| < 0.04$).

Motivation: Beyond Handcrafted Metrics

Current status: best single predictor ($S_{95,\mathrm{BP}}$) reaches only $|\rho_s| \approx 0.39$.

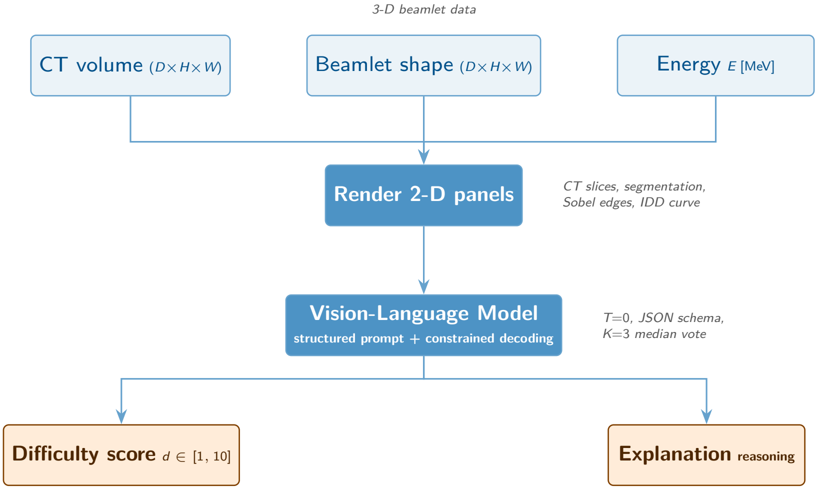
Key insight from Jungo et al. (2020) [1]: voxel-wise uncertainty is unreliable, but *subject-level aggregated* features accurately detect segmentation failures (AUC-ROC > 0.85).

Idea: replace manual feature engineering with a Vision-Language Model (VLM) that performs implicit visual aggregation over the beamlet geometry.

Core hypothesis

A VLM can capture richer spatial patterns (irregular bone-tissue interfaces, air cavities, complex anatomy near the beam path) than any single scalar metric, yielding a stronger and more *reproducible* difficulty score.

Proposed Pipeline: Overview



Input Specification

The VLM receives a **structured prompt** composed of:

1. **Image panel** (rendered from 3-D data):
   - Axial & coronal CT slices at BP depth
   - 7-class tissue segmentation overlay
   - Sobel edge-magnitude map (hot colourmap)
   - IDD curve with BP range annotation
   - Beamlet-shape projection (flux) overlay

2. **Text metadata**:
   - Initial beam energy E [MeV]
   - BP depth range $[z_{\min}, z_{\max}]$ [mm]
   - Number of density regions n_{reg}
   - Voxel resolution [mm]

Note: reuses the existing `plot_ct_with_segmentation` figures.

Limitation: ground-truth dependency

The current pipeline relies on **ground-truth dose** (MC) for BP range estimation, IDD rendering, and GPR labels, so it is not yet applicable in an **active learning** setting. Future work: replace GT-derived features with CT-only surrogates.

Reproducible Difficulty Scoring

Critical requirement: the scoring pipeline must be *deterministic and reproducible*.

Strategy: structured output with constrained decoding.

1. Set VLM temperature $T = 0$ (greedy decoding) to eliminate sampling randomness.
2. Use a JSON-schema-constrained output format (supported by OpenAI, vLLM, Outlines [2]): `{"difficulty": int, "confidence": float, "reason": str}`
3. Run K=3 independent forward passes per beamlet; report **median** score and inter-run agreement (κ) as a built-in reliability check.
4. Validate against held-out GPR ground truth with Spearman ρ and failure-detection AUC-ROC (following Jungo et al. [1]).

With constrained decoding and T=0, identical inputs produce identical outputs on the same model checkpoint.

Experimental Tracks

Track	Method	VLM input	Output
A	Zero-shot	Image + metadata prompt	Difficulty 1–10
B	Few-shot	Same + 5–10 labelled examples	Predicted GPR
C	Fine-tuned	LoRA adapter on VLM encoder	GPR (continuous)

Track A tests whether a general-purpose VLM can reason about dose-prediction difficulty *without any training*.
Track B adds in-context learning with a handful of extreme cases (best/worst GPR) as calibration anchors.
Track C fine-tunes a lightweight adapter (LoRA, Hu et al. [3]) on the n=3 950 dataset, mapping VLM visual features directly to continuous GPR.

Evaluation: Spearman/Pearson ρ vs. actual GPR, AUC-ROC for failure detection (GPR $<$ 95%), energy-stratified breakdown.

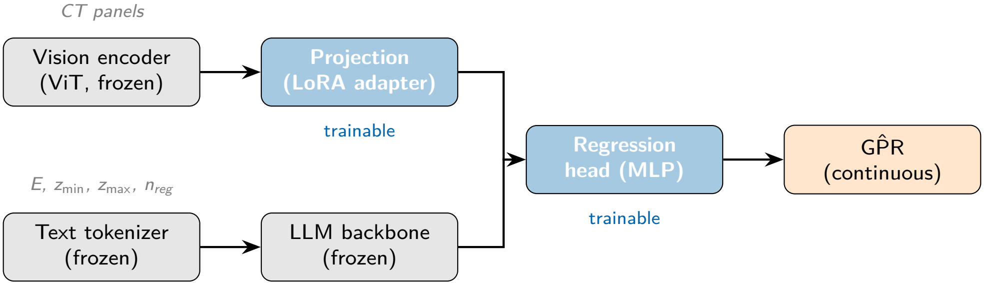
Computational Cost Analysis

Constraint: VLM scoring must be *significantly cheaper* than Monte Carlo simulation.

Method	Time/beamlet	GPU
MC simulation (TOPAS)	$\sim$60 s	none (CPU cluster)
ADoTA inference	1.7 ms	1$\times$ consumer GPU
VLM Track A (API)	$\sim$2 s	none (cloud)
VLM Track C (local, 7B)	$\sim$0.3 s	1$\times$ A100

- API-based Track A/B: $\sim$50$\times$ cheaper than MC.
- Local 7B model (Track C): $\sim$200$\times$ cheaper than MC, and all computation stays on-premise (data privacy).
- Full training plan with n=3 950: a single LoRA fine-tuning epoch on a 7B VLM takes $\sim$15 min on one A100 (Liu et al. [4]).

Track C: Fine-tuned VLM Architecture



Only the LoRA projection and the regression head are trained (∼2–4 M parameters). Vision encoder and LLM backbone stay frozen, making fine-tuning feasible on a single GPU.

Candidate models: LLaVA-1.5-7B [4], Qwen2-VL-7B [5], or BiomedCLIP [6] (if a medical-domain prior is desired).

References

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Next Steps

1. **Energy-specific models**: train or evaluate ADoTA separately on low- vs high-energy subsets to test whether performance gaps align with heterogeneity metrics.
2. **Composite difficulty score**: combine top metrics (e.g. $S_{95,\mathrm{BP}}$, σ_{BP}, $\Delta H/n_{\mathrm{reg}}$) into a single "difficulty index" via PCA or logistic regression.
3. **VLM difficulty estimation**: implement Track A (zero-shot) as a proof-of-concept on the n=3 950 dataset and compare with engineered metrics.
4. **Investigate low-energy worst cases**: visualise the beamlets with lowest GPR in the 70–100 MeV bin to identify anatomical patterns driving failure.

Questions?